

Resonance Warp in PGA–DST: Layer-Hopping vs On-Demand Phase Entry

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Abstract

We develop a hypothetical interstellar propulsion framework within the projective geometric algebra (PGA) extension of dual spacetime theory (DST). Gravity and inertia emerge from torsional mismatch between particle-intrinsic usual and dual spacetimes, lifted into $G(3, 1, 1)$ with a ten-dimensional bivector generator space (hyperbolic, cyclic, and null sectors). The extended motor $M = \exp(\Omega_{\text{biv}}^{(5)}) = RT$ is unitary ($M\widetilde{M} = 1$), preserving causal propagation on the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$. The torsion scalar J changes sign at odd multiples of π in the dual rotor angle θ_a , yielding repulsive layers ($J < 0$) that enable *effective* superluminal coasting without local causality violation.

Unlike Alcubierre-type metric engineering, warp in PGA–DST is not about bending a spacetime fabric with exotic matter; it is about coherently tuning collective dual-rotor resonance and entering $J < 0$ layers via circularly polarised THz driving. We quantitatively compare two operational regimes: (1) *layer-hopping*, exploiting slow cosmic phase drift of background dual rotors, and (2) *on-demand entry*, applying active phase driving. For million-tonne-class vessels at $0.1\text{--}0.5c$, layer-hopping reduces entry energy by 4–7 orders of magnitude, into the kWh–low MWh range. All derivations respect the $G(3, 1, 1)$ algebra, teleparallel equivalence with general relativity, and unitary motor causality. We connect repulsive $J < 0$ layers, helicity-controlled gravity engineering, layer stability, and an experimental roadmap. This paper is explicitly *hypothetical*: macroscopic dual-phase coherence remains unverified.

1 Introduction

Conventional warp-drive proposals, beginning with Alcubierre’s metric bubble [1], require exotic negative energy density and violate the null energy condition (NEC) [2]. Engineering such configurations at macroscopic scales appears prohibitive under known physics.

Dual spacetime theory (DST) offers a fundamentally different ontology [3]. There is no autonomous spacetime continuum; instead, *space emerges as collective resonance* among dual rotors carried by individual particles. Gravity is torsional mismatch between the usual and dual sectors, reformulated as teleparallel gravity (TEGR) [5] with vanishing curvature and non-zero torsion. The PGA extension [4] embeds this structure in projective geometric algebra $G(3, 1, 1)$, adjoining a null vector e_4 so that translations, light propagation, and torsion are unified in a single ten-dimensional bivector object propagated by a unitary motor.

In this framework, **warp propulsion** is not metric engineering but *resonance tuning*: shifting the collective dual-rotor phase to enter repulsive torsional layers ($J < 0$) where effective displacement exceeds the local light-speed bound, while information remains constrained to the light channel. This paper is a *hypothetical* extension of PGA–DST to interstellar propulsion. We preserve the quantitative energy estimates of the earlier DST warp analysis while grounding them in the PGA algebraic structure, local generation of repulsive $J < 0$ layers, and gravity-engineering mechanisms from the PGA textbook programme.

2 PGA–DST Foundations

2.1 Three-sector structure in $G(3, 1, 1)$

Each massive particle carries paired compactified Minkowski structures encoded in $\text{Cl}(3, 1) \cong \mathbb{H} \otimes \mathbb{H}$. The PGA lift adjoins e_4 with $e_4^2 = 0$ and $\{e_4, e_\mu\} = 0$, extending the algebra to $G(3, 1, 1)$ (32 real dimensions).

Vector generators are

$$e_0 = j, \quad e_1 = kI, \quad e_2 = kJ, \quad e_3 = kK,$$

with signature $(-, +, +, +)$. The bivector space has dimension ten and decomposes into three sectors [4]:

Sector	Generators	Square	Role
Hyperbolic	iI, iJ, iK	+1	Usual-spacetime boosts
Cyclic	I, J, K	-1	Dual-spacetime rotations
Null	$N_\mu = e_4 \wedge e_\mu$	0	Translations

The null sector realises the Poincaré translation ideal: $N_\mu N_\nu = 0$ for all μ, ν , and the Lie algebra closes as $\mathfrak{iso}(3, 1) = \mathfrak{so}(3, 1) \oplus \mathfrak{so}(3, 1) \ltimes \mathbb{R}^{3,1}$.

2.2 Rotors, motors, and the duality map

The usual and dual rotors are

$$R_{\text{usual}} = \exp\left(\sum_{a=1}^3 \frac{\phi_a}{2} i\Gamma_a\right), \quad R_{\text{dual}} = \exp\left(\sum_{a=1}^3 \frac{\theta_a}{2} \Gamma_a\right), \quad (1)$$

where $\Gamma_a \in \{I, J, K\}$. The relative mismatch rotor is

$$\Omega = R_{\text{usual}}^\dagger R_{\text{dual}}, \quad \Omega_{\text{biv}} = \log \Omega. \quad (2)$$

The duality map $X \mapsto Xi$ (with $i = e_0 e_1 e_2 e_3$) interchanges hyperbolic and cyclic sectors while leaving the null sector closed, since e_4 commutes with i .

The extended mismatch bivector and motor are

$$\Omega_{\text{biv}}^{(5)} = \underbrace{\sum_{a=1}^3 \left(\frac{\alpha_a}{2} B_a^+ + \frac{\beta_a}{2} B_a^- \right)}_{\Omega_{\text{torsion}}} + \underbrace{\sum_{\mu=0}^3 \frac{\lambda_\mu}{2} N_\mu}_{\Omega_{\text{trans}}} \quad (3)$$

$$M = \exp(\Omega_{\text{biv}}^{(5)}) = RT, \quad T = 1 + \sum_{\mu=0}^3 \frac{\lambda_\mu}{2} N_\mu, \quad (4)$$

where the null exponential truncates at first order because $N_\mu N_\nu = 0$.

Theorem 2.1 (Motor unitarity). *Every bivector generator is reversal-odd ($\tilde{B} = -B$). Consequently $R\tilde{R} = 1$, $T\tilde{T} = 1$, and $M\tilde{M} = 1$. The sandwich product $X' = MX\tilde{M}$ preserves the degenerate $G(3, 1, 1)$ metric and causal structure.*

2.3 Torsion scalar and gravitational action

The torsion scalar is defined via the Cartan–Killing form on the six-dimensional torsion sector:

$$J = \frac{1}{16} B_{\text{Killing}}(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = \frac{1}{2} \sum_{a=1}^3 (\alpha_a^2 - \beta_a^2). \quad (5)$$

The hyperbolic generators contribute $B(B_a^+) = +8$ and the cyclic generators $B(B_a^-) = -8$, so J changes sign when dual-sector rotation dominates. In the small-angle regime, with $\alpha_a \sim \sinh(\phi_a/2)$ and $\beta_a \sim \sin(\theta_a/2)$, (5) reduces to the finite-angle form

$$J \approx \sum_{a=1}^3 \left[\sinh^2\left(\frac{\phi_a}{2}\right) - \sin^2\left(\frac{\theta_a}{2}\right) \right], \quad (6)$$

used in the operational estimates below.

The PGA extended invariant couples torsion to translation:

$$J^{(5)} = J + \frac{1}{2}\eta_{\mu\nu}\lambda^\mu\lambda^\nu, \quad (7)$$

where $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$. The translational term vanishes on the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$, embedding null propagation algebraically.

Gravity follows from the action

$$S = \frac{c^4}{16\pi G} \int J \sqrt{-g} d^4x, \quad (8)$$

which is equivalent to the Einstein–Hilbert and TEGR actions: all GR solutions, including Schwarzschild, are recovered from tetrads generated by unitary motors, without invoking curvature [4, 5]. Interpretation changes; equations of motion do not.

3 What Warp Means in PGA–DST

3.1 Space as collective resonance

In PGA–DST, macroscopic space is not a pre-existing manifold but the statistical alignment of particle-intrinsic rotors across $\sim 10^{32}$ baryons per tonne of matter. The scalar J measures misalignment. When neighbouring dual rotors are aligned ($\theta_a \approx 2n\pi$), $J > 0$ and we experience attractive gravity. When driven to repulsive points ($\theta_a \approx (2n+1)\pi$), collective resonance shifts and effective geometry changes.

Warp propulsion takes two distinct operational meanings:

1. **Resonance-shifting warp (layer-hopping).** The vessel remains within the collective resonance manifold but shifts local dual-rotor phase to enter a repulsive layer ($J < 0$). The spacecraft rides a different harmonic of the same resonant structure. This is the primary mechanism analysed below.
2. **Subspace-transition warp (dual-sector immersion).** The vessel actively decouples from ambient collective resonance and immerses primarily in the dual sector. Re-emergence requires precise re-synchronisation with the target region’s phase. This is energetically far costlier than layer-hopping because it breaks macroscopic coherence entirely.

Thus warp is *changing the tuning of cosmic resonance*, not bending or tearing a fabric. The engineering challenge shifts from exotic negative energy to coherent dual-phase control at macroscopic scales.

3.2 Comparison with Alcubierre warp drives

Table 2: Alcubierre metric engineering vs. PGA–DST resonance warp (hypothetical)

Aspect	Alcubierre [1]	PGA–DST (this work)
Mechanism	Contracting/expanding spacetime bubble via $g_{\mu\nu}$	Collective dual-rotor phase tuning; $J < 0$ layers
Exotic matter	Required; violates NEC [2]	Not required; J sign from Killing-form asymmetry
Causality	Local c exceeded inside bubble	Effective superluminal <i>coast</i> ; local c preserved via $M\widetilde{M} = 1$
Information	Can outrun light within bubble	Bound to light cone; $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$
GR equivalence	Modified Einstein equations	TEGR/EH recovered; same weak-field limits
Engineering	Negative energy density at bubble wall	THz circularly polarised driving of θ_a

The contrast is ontological as well as operational. Alcubierre engineering deforms a background metric. PGA–DST has no background continuum; it modulates the torsion scalar J and, during coast, the translational component of $J^{(5)}$ on approximately light-like worldlines.

4 Repulsive Layers

4.1 Sign structure of J

Table 3: Torsion scalar sign and physical interpretation

Condition	Sign of J	Effect
$\alpha_a \gg \beta_a$ (aligned)	$J > 0$	Attractive gravity
$\alpha_a \approx \beta_a$	$J \approx 0$	Vacuum / shielding / inertia reduction
$\beta_a > \alpha_a$ (repulsive point)	$J < 0$	Repulsive gravity

Repulsive layers occur near $\theta_a = (2n + 1)\pi$. In strong-field astrophysical contexts, the same sign oscillation produces stratified “onion” structures (twist stars) rather than event horizons and singularities [3]: an infinite stack of alternating $J > 0$ and $J < 0$ shells provides the cosmological backdrop for layer-hopping.

In any region $\mathcal{R}$ with $J(x) < 0$, dual-sector rotation dominates, yielding effective repulsion. Such repulsive layers are engineerable in principle via coherent excitation of θ_a . For warp entry, a $J < 0$ shell is not merely a barrier but the *operational medium*: the vessel transitions into a locally generated or cosmically encountered repulsive layer, coasts with reduced gravitational coupling, and exits at a scheduled phase boundary. Stability under perturbation follows from self-reinforcing mismatch dynamics: intrusion of external matter increases $|\beta_a|$, driving J more negative.

5 Drive Mechanism: THz Helicity and Dual-Rotor Coupling

5.1 Electromagnetic splitting

The electromagnetic field splits across DST sectors:

$$F = F_{\text{usual}} + F_{\text{dual}}, \quad F_{\text{usual}} = \mathbf{E} \cdot (iI, iJ, iK), \quad F_{\text{dual}} = \mathbf{B} \cdot (I, J, K). \quad (9)$$

Electric fields couple to hyperbolic (usual) generators; magnetic fields couple exclusively to cyclic (dual) generators. This algebraic separation is the geometric origin of the distinct transformation properties under time reversal ($\mathbf{E} \rightarrow \mathbf{E}$, $\mathbf{B} \rightarrow -\mathbf{B}$).

5.2 Circular polarisation as a chiral switch

For a circularly polarised wave with helicity $\sigma = \pm 1$ propagating along $\hat{z}$,

$$\langle F_{\text{dual}} \rangle \propto \sigma E_0^2 K. \quad (10)$$

Helicity directly controls J :

- $\sigma = +1$ (right-handed): enhances attractive torsion ($J > 0$).
- $\sigma = -1$ (left-handed): reduces J ; sufficient intensity can invert to $J < 0$.

The resonance driving equation is

$$\dot{\theta} = \kappa \sigma E_0^2 \sin(\omega t - \theta + \delta_\sigma), \quad (11)$$

where κ is a matter-dependent coupling constant. Synchronising θ with ϕ drives $J \rightarrow 0$ (shielding, inertia reduction); driving θ past ϕ yields $J < 0$ (repulsion, warp entry).

5.3 THz collective resonance

Single-particle dual resonance lies at Planckian frequencies ($\sim 10^{43}$ Hz). Macroscopic coherence over $N \sim 10^{26}$ – 10^{32} particles redshifts the effective resonance to the THz band:

$$\omega_{\text{eff}} \sim \frac{\omega_{\text{Planck}}}{\sqrt{N}} \sim 10^{12}\text{--}10^{15} \text{ Hz}. \quad (12)$$

Superconducting phase locking and high- Q cavities further enhance coupling. This is the same mechanism proposed for table-top tests of $\Delta m/m \gtrsim 10^{-6}$ with helicity-reversal signatures [3]. Warp engineering is the macroscopic extrapolation of that laboratory programme.

6 Operational Regimes: On-Demand Entry vs Layer-Hopping

6.1 Cosmic phase drift

Large-scale structure induces slow background drift of dual-rotor phases:

$$\left| \frac{d\theta_a}{dt} \right| \simeq 1.1 \times 10^{-26} \text{ rad} \cdot \text{m}^{-1}, \quad \frac{d\theta_a}{dt} \simeq 3.3 \times 10^{-10} (v/c) \text{ rad} \cdot \text{s}^{-1}. \quad (13)$$

This continuously sweeps the infinite stack of repulsive layers past an observer moving at velocity v .

6.2 Effective Hamiltonian and entry energy

Near a repulsive layer, with $\theta = \theta_a - (2n + 1)\pi$,

$$H = \frac{p^2}{2m_{\text{eff}}} + V(\theta), \quad V(\theta) \approx \frac{1}{2}k\theta^2 - \frac{\hbar\omega_0}{2} \cos(\theta), \quad (14)$$

where m_{eff} is the collective dual excitation inertia and $\omega_0 \sim 10^{14}$ – 10^{15} Hz for THz collective modes. The semiclassical energy to reach phase offset $\Delta\phi$ is

$$E(\Delta\phi) \approx N_b \hbar\omega_{\text{res}} \left| \sin\left(\frac{\Delta\phi}{2}\right) \right|, \quad (15)$$

with $N_b \approx 6 \times 10^{32}$ baryons in a 10^6 kg vessel. Small $\Delta\phi$ gives quadratic scaling; near π the cost saturates, explaining the extreme sensitivity of on-demand entry.

During entry, the motor (4) is driven primarily through Ω_{torsion} toward $J < 0$. During coast in a repulsive layer, translational mismatch λ^μ is tuned to lie near the light cone so that the $J^{(5)}$ translational term is suppressed and displacement proceeds via phase-layer sliding rather than acausal local boosts.

6.3 On-demand entry energy

Worst-case on-demand entry ($\Delta\phi_{\text{max}} \approx \pi$) for a 10^6 kg vessel:

Table 4: On-demand subspace entry (worst-case $\Delta\phi \simeq \pi$)

Cruise velocity	γ	Energy required (10^6 kg vessel)
$0.10c$	1.005	$\sim 8 \times 10^{11} \text{ J} \approx 220 \text{ GWh}$
$0.30c$	1.048	$\sim 7.5 \times 10^{11} \text{ J} \approx 210 \text{ GWh}$
$0.50c$	1.155	$\sim 6.9 \times 10^{11} \text{ J} \approx 190 \text{ GWh}$
$0.99c$	~ 7	$\sim 3 \times 10^{11} \text{ J} \approx 80 \text{ GWh}$

These values approach projected compact fusion or antimatter-catalysed limits, making pure on-demand entry technologically challenging.

6.4 Layer-hopping performance

Waiting time for phase deficit below threshold $\Delta\phi_{\text{threshold}}$:

$$\tau \simeq \frac{\Delta\phi_{\text{threshold}}}{|d\theta/dt|} = 3 \times 10^9 \Delta\phi_{\text{threshold}} \left(\frac{c}{v} \right) \text{ s}. \quad (16)$$

A threshold $\Delta\phi \approx 10^{-3}$ offers a practical compromise between waiting time and power. For a 10 light-year mission, total energy expenditure drops to 10–100 MWh under the hybrid protocol.

7 Causality, Horizons, and Effective Superluminal Coasting

7.1 Unitary motors and local causality

Theorem 2.1 guarantees that every physical transformation is implemented by a unitary motor $M\widetilde{M} = 1$. No sandwich product can produce acausal local propagation: the light cone $\eta_{\mu\nu}\lambda^\mu\lambda^\nu = 0$ is built into $J^{(5)}$ (7). Photons correspond to full anti-synchronisation ($\phi = \theta$, $\Omega = 1$) with vanishing translational contribution in $J^{(5)}$.

Table 5: Layer-hopping performance at $v = 0.3c$ for a 10^6 kg vessel

Threshold $\Delta\phi$	Typical wait	Energy required	Power (10-min pulse)	Gain
π (on-demand)	0	750 GWh	—	$1\times$
10^{-1}	~ 1 yr	75 GWh	450 MW	$\sim 10^4\times$
10^{-2}	~ 40 days	7.5 GWh	45 MW	$\sim 10^5\times$
10^{-3}	~ 4 days	750 MWh	4.5 MW	$\sim 10^6\times$
10^{-4}	~ 10 hrs	75 MWh	450 kW	$\sim 10^7\times$
10^{-5}	~ 1 hr	7.5 MWh	45 kW	$\sim 10^8\times$

7.2 Effective superluminal displacement

“Effective superluminal coasting” means that the *phase-layer displacement rate* of the vessel exceeds c as measured against the ambient $J > 0$ reference frame, while:

1. local signal velocity remains $\leq c$ in the vessel’s instantaneous frame;
2. information about the hop is carried only on the light channel;
3. no closed timelike curves are generated because hops are finite-duration transitions between globally hyperbolic TEGR patches.

This is analogous to a refractive index change in a medium: phase velocity can exceed c without violating relativity, provided group velocity and information velocity do not.

7.3 Horizons and quasi-horizons

PGA–DST excludes true event horizons and central singularities in favour of stratified twist-star interiors [3]. However, *quasi-horizons*—regions where $\int J dr$ rises steeply—can impede exit without supplying sufficient torsion energy. Warp navigation must map local phase gradients to avoid trapping in quasi-horizon shells. This is consistent with the layer-stability analysis below.

8 Mission Protocol, Stability, and Navigation

8.1 Hybrid interstellar protocol

1. Cruise at $0.2\text{--}0.5c$ using nuclear or beamed propulsion.
2. Monitor local dual phase via a compact THz resonant cavity coupled to dual degrees of freedom.
3. When $\Delta\phi < 10^{-3}$ rad (every few days to weeks at $0.3c$), apply a low-power, left-handed circularly polarised THz pulse to complete the transition into $J < 0$.
4. Coast in the repulsive layer for weeks to decades, achieving effective superluminal displacement.
5. Exit via complementary pulse (helicity reversal or phase realignment) and repeat, scheduling hops from phase-gradient maps.

8.2 Layer stability

Inside a repulsive layer, small deviations $\delta\theta$ experience restoring forces from $V(\theta)$ in (14). Characteristic oscillation frequency:

$$\omega_{\text{stab}} \approx \sqrt{\frac{k}{m_{\text{eff}}}}. \quad (17)$$

Active feedback via the same THz cavity damps unwanted excitations. Self-reinforcing mismatch dynamics in $J < 0$ regions provide additional stability against external perturbation.

8.3 Navigation in torsional phase space

Navigation requires control of entry/exit points and hop direction. We propose auxiliary low-mass probes to map local θ_a gradients, enabling predictive scheduling analogous to gravitational slingshots but in torsional phase space. Motor factorisation $M = RT$ separates rotational mismatch (layer selection via R) from translational coast (via light-like T).

9 Experimental Roadmap and Open Questions

9.1 Near-term laboratory tests

Before any warp application, the foundational hypothesis—macroscopic dual-phase coherence coupled to EM fields—must be tested:

1. **Table-top gravity engineering.** Superconducting test masses in high- Q THz cavities; monitor $\Delta m/m \gtrsim 10^{-6}$ with helicity-reversal sign change [3].
2. **Repulsive-layer generation.** High-intensity circularly polarised THz (10^{12} – 10^{15} Hz) to drive $J < 0$ locally; measure repulsive deflection or weight reduction.
3. **Resonance spectroscopy.** Systematic frequency/helicity scans to constrain κ and compactification scale r_{comp} .

9.2 Astronomical and space-based tests

1. Precision torsion balances or atom interferometers to detect cosmic phase drift (13).
2. Satellite baselines to measure large-scale dual-phase gradients across the solar system.
3. Correlation of repulsive-layer signatures with galactic-scale $J < 0$ boundaries in rotation-curve data [3].

9.3 Key open questions

- What is the precise coupling κ between EM fields and dual rotors?
- What is the decoherence time for macroscopic dual-phase coherence?
- Can collective $N \sim 10^{32}$ baryon locking be maintained during a hop pulse?
- How do quasi-horizon shells interact with repeated layer transitions?

10 Discussion and Conclusions

We have upgraded the dual spacetime warp propulsion hypothesis to the PGA–DST framework. The central claims are:

1. Warp is resonance tuning of collective dual rotors, not Alcubierre metric engineering.
2. Repulsive layers ($J < 0$) provide the operational medium for effective superluminal coasting.
3. THz circularly polarised driving with helicity $\sigma = -1$ is the proposed entry mechanism.
4. Layer-hopping exploits cosmic phase drift to reduce entry energy by 4–7 orders of magnitude relative to on-demand entry.
5. Motor unitarity $M\widetilde{M} = 1$ and the light-cone structure of $J^{(5)}$ preserve local causality.

All quantitative estimates inherit the semiclassical model (15) and remain *hypothetical* until macroscopic dual-phase coherence is experimentally confirmed. Failure of table-top tests would constrain κ and r_{comp} without falsifying the algebraic PGA–DST structure, which independently recovers GR via TEGR. Success at laboratory scales would open a path toward engineering repulsive layers and, in principle, interstellar layer-hopping.

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